

Greedy Pig

Purpose:

This activity provides students with a fun game context in which to practice their addition skills. It also introduces concepts of probability.

Achievement Objectives:

S4-3: Investigate situations that involve elements of chance by comparing experimental distributions with expectations from models of the possible outcomes, acknowledging variation and independence.
[AO elaboration and other teaching resources](#)

S4-4: Use simple fractions and percentages to describe probabilities.
[AO elaboration and other teaching resources](#)

Specific Learning Outcomes:

- Perform addition calculations using numbers 1-6

Description of mathematics:

The mathematics in this unit is at the higher end of Level 4, but should be manageable with support from the teacher. Students will need to understand how to use powers of numbers in practical situations, and how to draw and interpret probability trees. This unit investigates expected outcomes of probability events, in the practical context of the game Greedy Pig.

Required Resource Materials:

dice
paper and pencils for recording

Key Vocabulary:

expected outcomes, experimentation, theoretical probability, probability tree, average

Activity:

Session 1

In this introductory session we introduce the game of Greedy Pig and play it several times to give students a chance to identify some possible strategies.

1. Introduce the game of Greedy Pig and explain the rules.

- Greedy Pig
- i. To play this game you need an ordinary 6-sided die.
 - ii. Each turn of the game consists of one or more rolls of the die. You keep rolling until you decide to stop, or until you roll a 1. You may choose to stop at any time.
 - iii. If you roll a 1, your score for that turn is 0.

- iv. If you choose to stop rolling before you roll a 1, your score is the sum of all the numbers you rolled on that turn.
 - v. The player with the highest score wins.
 - vi. Each player has 10 turns.
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2. Ensure that students understand the rules. Get someone to explain the rules in their own words. A common misunderstanding is the difference between a turn and a roll of the die.
 3. Introduce a variation to the game. Instead of individual students rolling the die on their turn the teacher will roll the die and all students will play at the same time. Everyone continues to add to their score until they choose to stop, at which point they can record their score for that turn. Students can remain 'in' for as long as they wish. If a 1 is rolled anyone who is still in gets a 0 for that turn and the next turn starts.
 4. Play the game a few times so that students see how it works.
 5. Ask students to describe how they decide when to save their score. Ask them to justify why they use their strategy. Discourage responses such as "If I feel lucky I roll again".
 6. List some of the strategies.
 7. Compare how successful some of the strategies are.
 8. Vote as a class for what students think is the best strategy.

Session 2

In this session we investigate some of the strategies to see which are more successful.

1. Play a couple of games of Greedy Pig as a class, with students using whatever strategy they prefer.
2. Explain that we are going to try to find out what strategy will give the highest score.
3. Ask students to suggest how to determine which the best strategy is.
4. It is likely that they will suggest trying different strategies and seeing which gives the highest score. Try this approach first.
5. Get students to choose a strategy that they will use to determine whether to continue or whether to save their score.
6. Ask students to tell you what strategy they will use.
7. Ensure that all strategies are specific, ie. that anyone using that strategy would play the same. It is likely that most students will choose strategies based around either numbers of rolls (ie. Stay in until there have been 4 rolls and then save your score.) or around obtaining a total for the turn (ie. Stay in until you have 15 points and then save your score).
8. Explain that we are going to record the total for each turn rather than the total for each game, as the best strategy should give the best average score for turns in the long run.
9. Play 20 turns asking each student to keep track of their score for each turn
10. Compare results.
Who has scored the most for the 20 turns?
What strategy did they use? What was their average score per turn?
11. Ask all students to work out their own average score per turn.
12. Discuss why some strategies are better than others.

Sessions 3-4

In these two sessions we try to find the best number of rolls of the die to remain in the game by experimentation and by theoretical probability. There is a lot to cover in this sequence, and will need to be split into at least two sessions. Where the split occurs depends on the ability of the students. For some classes students can be expected to work out much of the theoretical understandings themselves; for others – a lot of support will be required. Revision of aspects of theoretical probability and probability trees may be required.

1. Play a game of Greedy Pig, with students choosing a maximum number of rolls of the die to remain in the game for. (ie. Stay in until there have been four rolls and then save your score.)
2. Compare the success of different maximum numbers of rolls.
Which was most successful?
How much did each number score on average? (Remember to include the zeroes when calculating averages.)
3. Ask students whether they can tell you for certain what the best number of rolls to stay in for is. Hopefully they will realise that with only 10 turns tested they can not know for sure which number of rolls will be most successful in the long run.
4. Ask for suggestions as to how you could find the best number of rolls to stay in for.
5. Try a larger sample size. Students will no doubt have realised that they need to stay in for at least three rolls before they get good scores, and will also have noticed that staying in for more than around 8 rolls is likely to result in almost all zero scores
6. Try playing 20 or 30 turns with students trying the strategy of staying in for numbers of rolls within this range.
7. Discuss the results. Ask students whether they now know for certain what is the best number of rolls to stay in for. They will no doubt have strong opinions as to how many rolls they should stay in for but should realise that there is still a random element.
8. At this stage you should suggest a theoretical approach, using probabilities.
For any given roll, how many points will you gain on average?

This may need to be elaborated on if they do not understand:

What are the possible results of staying in for one more roll at any time?

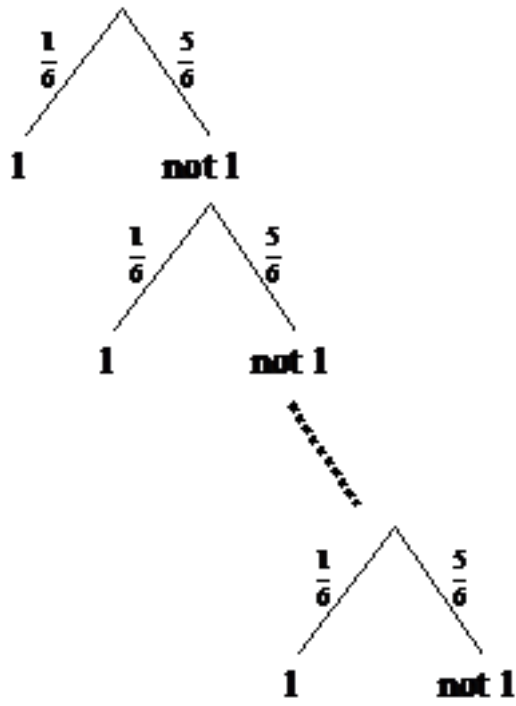
The possibilities are – they could roll a 1 and hence lose their score for the turn, or they could gain 2, 3, 4, 5, or 6 points.

So, for the first roll, the expected score is the average of the 6 possible scores, or $(0+2+3+4+5+6)/6$, which equals $20/6$, or $3\frac{1}{3}$. If you stay in for one roll only, you will get $3\frac{1}{3}$ on average per turn.

9. How does this work for staying in for more than one roll?

If you elect to stay in for two rolls, there is still a $1/6$ chance that you will roll a 1 on the first roll, in which case your total score for the turn is 0.

There is also a $1/6$ chance that you will roll a 1 on the second roll, in which case your total score is 0. Maybe it would help to draw a probability tree of this. A probability tree with all 6 possible outcomes (1-6 on the die) will get too big very quickly, so lets just look at the ‘big’ possibilities, which are getting a 1, or not getting a 1. This will at least help us work out the probability of getting a total score of 0.



1. So how do we work out the probability of getting 0 from this? Hopefully students will be able to see that for any given number of rolls there is a $(5/6)^n$ chance of getting a score, as all the other branches end in 1, or a score of 0. Therefore, for any given number of rolls, there is a $1 - (5/6)^n$ chance of getting 0.
2. This is useful, at least we know what the chance of scoring nothing is for any chosen number of rolls, and we can see that the chance of getting 0 goes up the more rolls we stay in for. But the more rolls we stay in for the more points we will score if we don’t roll a 1. How can we work out how much we are likely to score?
3. If we do not roll a 1, there are five possibilities, each of which will add to our score. *What is the average gain from these possibilities?* This can be found by simply averaging the five values. $(2+3+4+5+6)/5 = 4$. So if we don’t roll a 1 and lose our whole score, we will on average gain 4 points per roll.
4. Applying this new knowledge to the probability tree above tells us that for any given number of rolls chosen we have a $1 - (5/6)^n$ chance of getting 0 and a $(5/6)^n$ chance of getting a score, which will on average be $4n$.
5. So the expected score for n rolls is $(1 - (5/6)^n) \times 0 + ((5/6)^n) \times 4n$
And since $(1 - (5/6)^n) \times 0 = 0$ we can simplify this to:
The expected score for n rolls is $(5/6)^n \times 4n$.
6. Have students make a table of the expected scores for using a strategy of staying in the game for 1 – 10 rolls (you could use Excel or similar if you wish).

Number of rolls	Chance of scoring	Score	Expected total score
n	$(5/6)^n$	4n	$(5/6)^n \times 4n$
1	0.83	4	3.33
2	0.69	8	5.56
3	0.58	12	6.94
4	0.48	16	7.72
5	0.40	20	8.04
6	0.33	24	8.04

7	0.28	28	7.81
8	0.23	32	7.44
9	0.19	36	6.98
10	0.16	40	6.46

Note: All values to 2 decimal places

- So if you are going to adopt a strategy of staying in for the same number of rolls each turn, you should stay in for either 5 or 6, in which case you will get an average score of just over 8 per turn.
- Discuss problems with adopting a strategy of staying in for a certain number of rolls. The biggest problem with this strategy is that if you roll 4 sixes in a row you are obliged to roll again and risk your score of 24, and conversely, if you roll five twos in a row you should stop on only 10 for the turn.

Session 5.

In today’s session we look at the alternative strategy of choosing a certain target score and stopping if you reach it on any given turn.

- Play a game of Greedy Pig, with students choosing a target total for each turn. (ie. Stay in until you have 15 points and then save your score)
- Compare the success of different target totals.
Which was most successful?
How much did each number score on average? (Remember to include the zeroes when calculating averages.)
- Ask students whether they can tell you for certain what the best target total is. Hopefully they will realise that with only 10 turns tested they can not know for sure which target total will be most successful in the long run. They will probably have realise that they should stay in until they have at least 10, but that 25 is too good a score to risk.
- Ask students how we could find the best target total. Hopefully they will have realised from the previous sessions that a theoretical approach is more accurate.
- How can we find what scores you should remain in on, and which you should save? Ask the class for suggestions.
- Again we want to look at the two possibilities: either we roll a 1 and lose our whole score, or we roll a different number in which case we expect to gain on average 4 points.
- So what are the expected outcomes for each possible value. If we have 0 points then there is a 1/6 chance that we will roll a 1 and get 0, and a 5/6 chance that we will get more points (an average of 4), so the expected outcome is (1/6 x 0)+(5/6 x 4) = 3.33.
- Have students make a table of the expected scores for using a strategy of staying in the game until you have a total of at least 10 - 25 (you could use Excel or similar if you wish).

Total	Total if 1 not rolled	Expected total
n	n+4	5/6(n+4)
10	14	11.67
11	15	12.50
12	16	13.33
13	17	14.17
14	18	15.00

15	19	15.83
16	20	16.67
17	21	17.50
18	22	18.33
19	23	19.17
20	24	20.00
21	25	20.83
22	26	21.67
23	27	22.50
24	28	23.33
25	29	24.17

Note: All values to 2 decimal places

- Ask students to explain what this means.
- Hopefully some will realise that although the expected totals keep going up, once they reach 20, the expected total for rolling again is lower than the score if they save their score where they are. Therefore, if your score is less than 20 you should roll again, but if it is more than 20 you should save your score.
- Now ask the students :
What is the best strategy for playing Greedy Pig?
- Discuss whether choosing a number of rolls to stay in for or choosing a target total is a better strategy.

Related Resources:

Learning Objects

New Zealand teachers can access a collection of online learning objects in the Digistore. This includes learning objects focused on probability. Of particular relevance to this unit are the learning objects called [Dice Duels](#) and [Coin Tossing](#) as the concepts explored are similar. Click here for [information about using the Digistore](#).

Figure it Out

Some links from the Figure It Out series which you may find useful are:

Statistics, Level 3-4, Picking Pocket Money, page 18

Statistics, Level 3-4, Left to Chance, page 21

Statistics, Level 3-4, Digital dilemma, page 22

Statistics, Level 4, Across the River, page 19

Statistics, Level 4, Wallowing Whales, pages 20 21 Level 4, Dodgy Dice, page 22

Add this resource to one of your plans: